1

- Stabilization: is the process of blending and mixing materials with a soil to improve certain properties. The process may include the blending of soils to achieve a desired gradation or the mixing of commercially available additives that may alter the gradation, texture, plasticity, or act at a binder for cementation of the soil.

2

Road Design Factors

- Desired life expectancy
- Expected traffic loading
- Surface characteristics/requirements
- Climate and topography
  - Drainage
  - Freeze/Thaw
  - Sub-grade geology

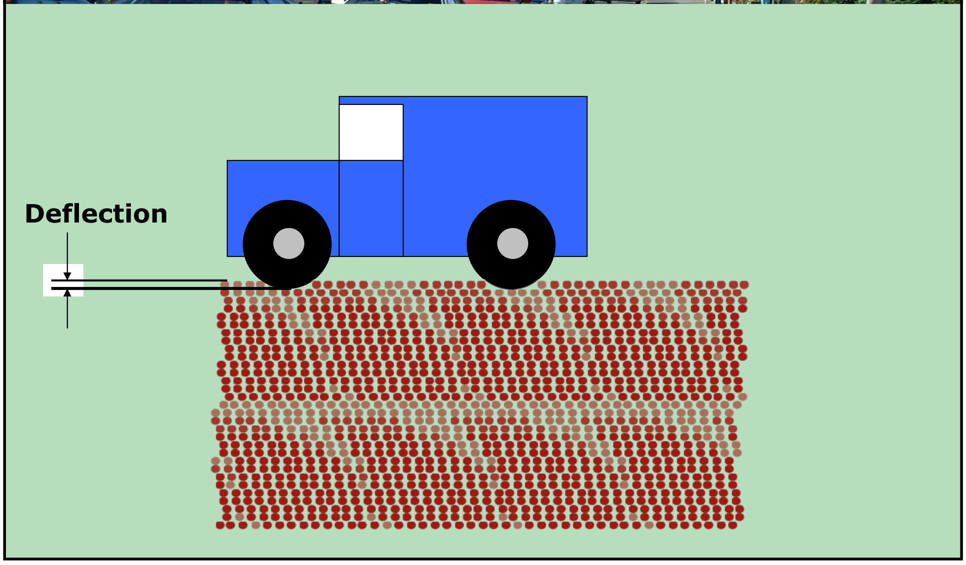
3

Soil Problems

- In-place soils may not have desired characteristics
  - Low in strength
  - High plastic fines content
  - Inadequate drainage or excessive capillarity
  - Unsuitable sub-grade resistance
  - Low resistance to surface wear

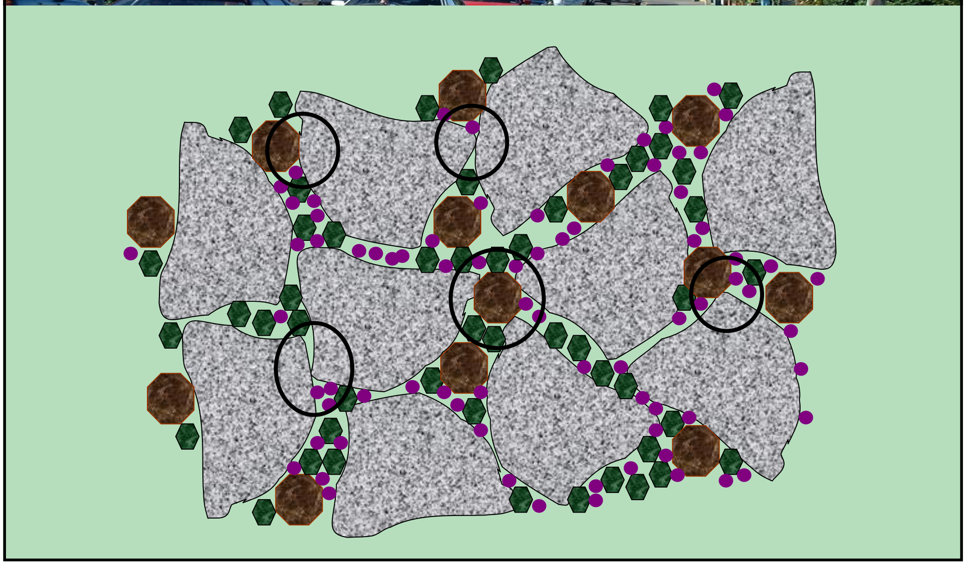
4

-Soil structures move or *deflect* under axle loads



5

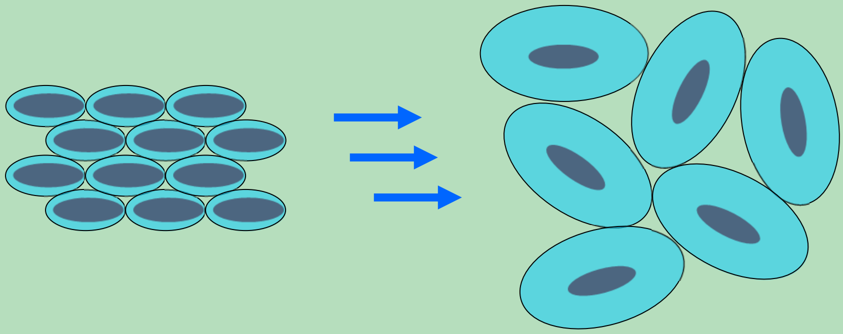
Bridging between larger soil particles (gravel) gives strength or stiffness to road structures



6

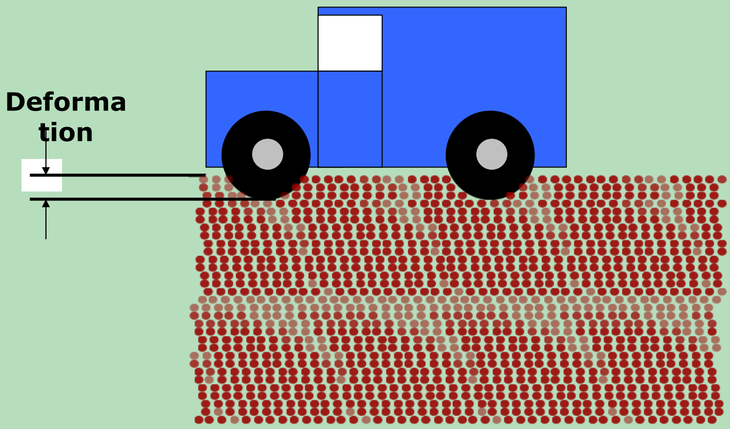
Water Destabilises “Fine-grained” Soils

- Water binds to small soil particles
- Water expands and lubricates the soil structure
- Destabilizing effects increase with % water in soil
- Destabilizing effects increase with % “fines” in soil



7

Water penetration causes “fine grained” soils” to *liquefy* and *deform.*



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10

11

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Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

Stabilising soil structures means <u>limiting</u> the <u>movement</u> of soil particles under the weight of axle wheel loads by improving soil stiffness and shear strength.

12

Stabilisation Approaches

- Historical - replace inadequate soils with select materials
- Historical - place new select materials onto the existing surface
- New - improve economically the engineering properties of in-situ or available materials

13

Traditional Cost Constraints

- Excavation, removal and disposal of unsuitable materials and surfaces
- Purchase and hauling of select materials
- Equipment capacity and machine time
- Length of project schedule
- Accessibility to existing infrastructure
- Funding limitations

14

15

Types of Stabilisation

- Mechanical Stabilisation: is accomplished by mixing or blending soils of two or more gradations to obtain a material meeting the required specifications.
- Soil blending may occur at the construction site, central plant, or a borrow area.

16

Modifications versus Stabilisation

- Additive stabilisation: accomplished by the addition of proper percentages of cement, lime, fly ash, bitumen, or combinations of these materials to the soil.
- Modifications: refers to stabilisation process that results in improvement in some property of the soil but does not result in a significant increase in soil strength and durability.

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Uses of stabilisation

- Quality Improvement
  - Better soil gradation
  - Reduction of plasticity index or swelling potential
  - Increase in durability
  - Increase in Strength
- Thickness Reduction
  - The improved strength and stiffness can result in a reduced thickness.

18

Selection of Stabilisation Type

- Factors to be considered
  - Type of soil
  - Purpose of stabilization
  - Required strength and durability
  - Environmental conditions
  - Cost

  There may be more than one candidate stabiliser for one soil type, however, there are some general guidelines that make specific stabiliser more desirable based on the soil granularity, plasticity, or texture.

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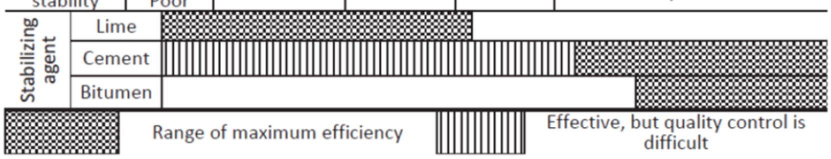
Type of preferred stabiliser based on the type of soil

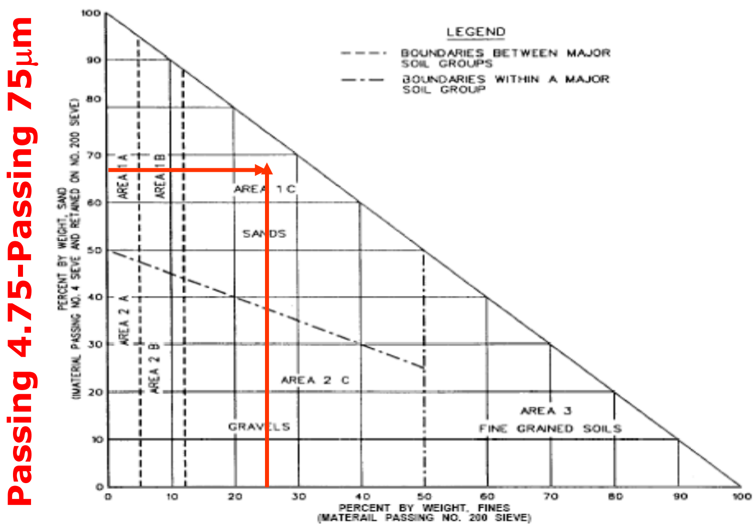
Designation	Fine clays	Coarse clays	Fine silts	Coarse silts	Fine sands	Coarse sand	Aggregate
Particle size (mm)	<0.0006	0.0006-0.002	0.002-0.01	0.01-0.06	0.06-0.4	0.4-2.0	>2.0
Volume stability	Very Poor	Fair	Fair	Good	Very Good		



20

- Examples:
- Portland cement: needs sufficient fines and high plastic materials should be avoided.
- Lime: will react with soils of medium to high plasticity to produce decreased plasticity, increased workability, reduced swell, and increased strength.

21



Passing 75µm Sieve

22

Area	Soil Class.[a]	Type of Stabilizing Additive Recommended	Restriction on LL and PI of Soil	Restriction on Percent Passing No. 200 Sieve[a]	Remarks
1A	SW or SP	(1) Bituminous			
		(2) Portland cement			
		(3) Lime-cement-fly ash	PI not to exceed 25		
1B	SW-SM or SP-SM or SW-SC or SP-SC	(1) Bituminous	PI not to exceed 10		
		(2) Portland cement	PI not to exceed 30		
		(3) Lime	PI not to exceed 12		
		(4) Lime-cement-fly ash	PI not to exceed 25		
1C	SM or SC or SM-SC	(1) Bituminous	PI not to exceed 10	Not to exceed 30% by weight	
		(2) Portland cement	--[b]		
		(3) Lime	PI not less than 12		
		(4) Lime-cement-fly ash	PI not to exceed 25		
2A	GW or GP	(1) Bituminous			Well-graded material only
		(2) Portland cement			Material should contain at least 45% by weight of material passing No. 4 sieve
		(3) Lime-cement-fly ash	PI not to exceed 25		
2B	GW-GM or GP-GM or GW-GC or GP-GC	(1) Bituminous	PI not to exceed 10		Well-graded material only
		(2) Portland cement	PI not to exceed 30		Material should contain at least 45% by weight of material passing No. 4 sieve
		(3) Lime	PI not less than 12		
		(4) Lime-cement-fly ash	PI not to exceed 25		
2C	GM or GC or GM-GC	(1) Bituminous	PI not to exceed 10	Not to exceed 30% by weight	Well-graded material only
		(2) Portland cement	--[b]		Material should contain at least 45% by weight of material passing No. 4 sieve
		(3) Lime	PI not less than 12		
		(4) Lime-cement-fly ash	PI not to exceed 25		
3	CH or CL or MH or ML or OH or OL or ML-CL	(1) Portland	LL less than 40 and PI less than 20		Organic and strongly acid soils falling within this area are not susceptible to stabilization by ordinary means
		(2) Lime	PI not less than 12		

[a] Soil classification corresponds to MIL-STD-619B. Restriction on liquid (LL) and plasticity index (PI) is in accordance with Method 103 in MIL-STD-621A.

[b] $PI \leq 20 + \frac{50 - \text{percent passing No. 200 sieve}}{4}$

23

Strength Requirements

Minimum Unconfined Compressive Strength for cement, lime, lime-cement fly ash stabilised soils.		
Stabilised Soil Layer	Flexible Pavement	Rigid Pavement
Base Course	5,171 kPa (750 psi)	3,447 kPa (500 psi)
Subbase Course, Select material or Subgrade	1,723 kPa (250 psi)	1379 kPa (200 psi)

24

Durability Requirements

Type of Soil Stabilised	Max. Allowable Weight Loss after 12 Wet-Dry or Freeze Thaw cycles
Granular, PI<10	11%
Granular, PI>10	8.0%
Silt	8.0%
Clays	6.0%

25

- Example:
- Passing #4 = 93%
- Passing # 200 = 25%
- PI=11
- Passing #4 and retained on # 200 =93-25= 68%
- Bituminous stabilization is not acceptable (PI>10)
- Portland cement is a candidate.
- Lime is not a candidate (PI<12)
- LCF is also a candidate.

26

Construction cut-off dates

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- Materials stabilized with cement, lime, or LCF should be constructed early enough during the construction season to allow the development of adequate strength before the first freezing cycle begins.
- The rate of strength gain is substantially lower at 10 °C than at 20 or 27 °C.
- Chemical reactions will not occur rapidly for lime stabilized soils when the soil temperature is less than 15 °C.

27

Determination of Stabiliser Content

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Lime stabilisation:

- All lime treated fine grain soils exhibit decreased plasticity, improved workability and reduced volume change characteristics.
- Not all soils exhibit improved strength characteristics.

28

Factors affecting Lime Stabilisation

- Soil type
- Lime type
  - Hydrated high calcium lime
  - Monohydrated dolomitic lime
  - Calcitic quicklime
  - Dolomitic quicklime
- Lime percentage
- Curing conditions (time, temperature, moisture)

29

Gradation Requirements for Lime Stabilised Base & Subbase courses

Type of Layer	Sieve Size	% Passing
Base	37.5 (1.5")	100
	19 (3/4")	70-100
	4.75 (#4)	45-70
	0.42 (#40)	10-40
	0.075 (#200)	0-20

Type of Layer	Sieve Size	% Passing
Subbase	37.5	100
	19	45-100
	0.42	10-50
	0.075	0-20

30

Lime Stabilisation

- Free lime (CaO) reacts with the pozzolans (Al_2O_3, SiO_2, Fe_2O_3) in the presence of water.
- The hydrated calcium silicate gel or calcium aluminate gel (cementitious material) can bind inert material together. For lime stabilization of soils, pozzolanic reactions depend on the siliceous and aluminous materials provided by the soil. The pozzolanic reactions are as follows:

31

Pozzolanic Reactions

$Ca(OH)_2$ => Ca++ + 2[OH]-

Ca++ + 2[OH]- + SiO_2 => CSH (Silica Gel)

Ca++ + 2[OH]- + Al_2O_3 => CAH (Alumina Gel)

The rapid rate at which hydration of the tricalcium aluminate occurs results in the rapid set of these materials, and is the reason why delays in compaction result in lower strengths of the stabilized materials.

32

Lime Content for Soil Modification

- The amount of lime is determined by trial and error.
- Reducing soil plasticity for example or Swell Potential:
  - Prepare successive samples of soil-lime mixtures at different treatment levels of lime.
  - Measure the PI for the different soil mixtures.
  - The minimum lime content that yields the desired PI is selected.
  - For swelling the lowest lime content that reduce the swelling potential is determined.

33

Lime content for lime stabilized soils

- pH Method:
  - Several lime soil slurries are prepared at different lime treatment levels (2, 4,6 & 8%)
  - pH value of each slurry is determined.
  - The lowest lime content at which a pH value of about 12.4 (the pH of free lime) is obtained.
  - This value is the initial design lime content.
- Alternate Method
  - use of attached chart

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Free lime at Ph Value>12.4

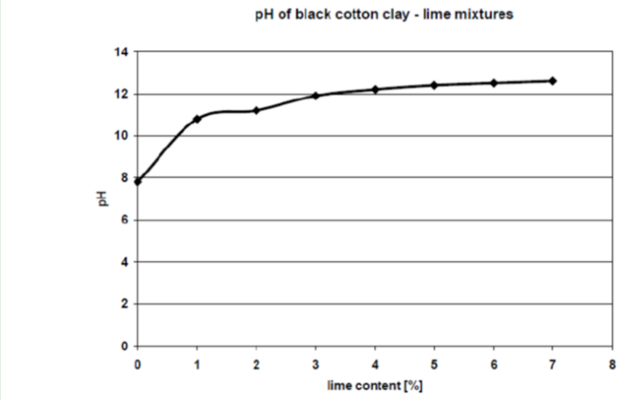
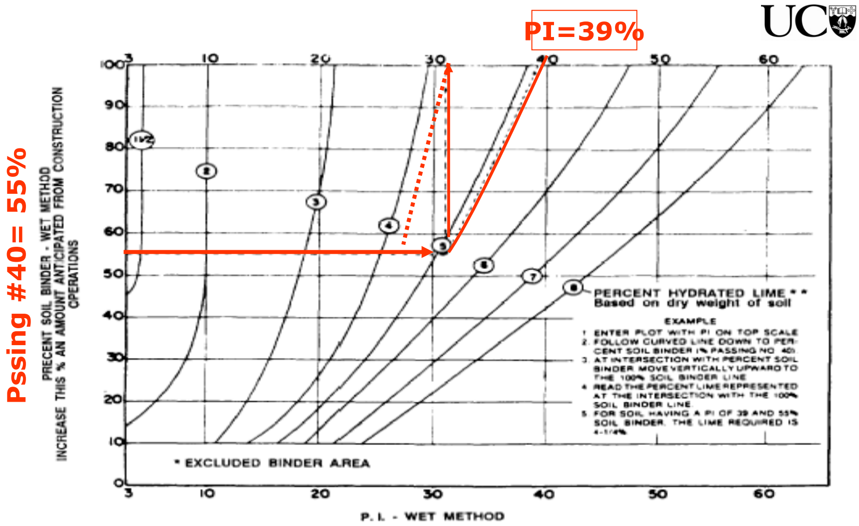


Figure 247: pH of Black Cotton Clay – lime mixtures.

35

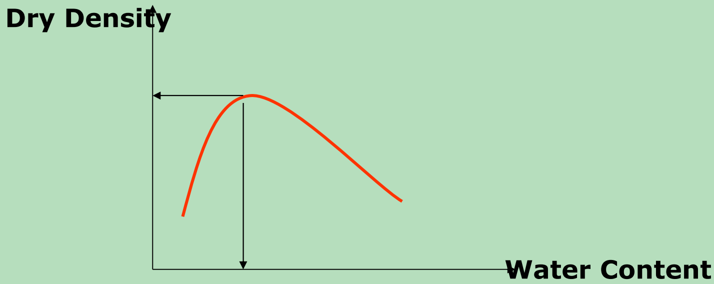


* Exclude use of chart for materials with less than 10% - No. 40 and cohesionless materials (P. I. less than 3)

* * Percent of relatively pure lime usually 90% or more of Ca and/or Mg hydroxides and 85% or more of which pass the No. 200 sieve. Percentages shown are for stabilizing subgrades and base courses where lasting effects are desired. Satisfactory temporary results are sometimes obtained by the use of as little as 1/2 of above percentages. Reference to cementing strength is implied when such termes as "Lasting Effects" and "Temporary Resulst" are used.

36

- Using the initial lime content, conduct the moisture density tests to determine the optimum moisture content and the maximum dry density.



37

The effect of the lime treatment on the CBR is shown in table 53. The samples were prepared at an initial moisture content of around 40%.

	0 days curing	7 days curing	28 days curing
0% lime	4.5		
5% lime	12	27	27
7% lime	20	35	39

Table 53: Effect of lime treatment on the CBR value [%].

38

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- Prepare Triplicate samples of the soil lime mixture for unconfined compression test and durability test. If the pH value test was used, prepared samples at the initial lime content and 2:0% and 4:0% above.
- If the alternate method is used, prepare samples at ±2.0% of the initial design lime content.
- Check the UCS and durability against the required criteria.

39

Cement Stabilization

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Unified Soil Classification	Initial Estimated Cement Content (Percent by dry weight)
GW, SW	5
GP, GW-GC, GW-GM, SW-SC, SW-SM	6
GC, GM, GP-GC, GP-GM, GM-GC, SC, SM, SP-SC, SP-SM, SM-SC, SP	7
CL, ML, MH	9
CH	11

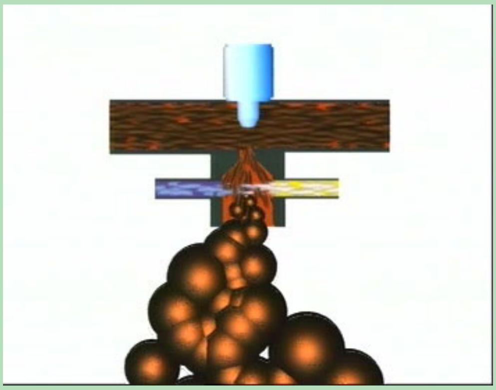
40

- Based on the soil classification determine the initial design cement content.
- Using the initial cement content, conduct the moisture density test and determine the optimum moisture content and maximum density.
- Prepare triplicate samples at the initial design cement content ± 2.0%
- Test UCS and durability test.

41

Foamed Bitumen Stabilisation

2 – 3 % water in 180 °C hot bitumen:
The bitumen expands 15 to 20 times its original volume.



The increased surface area makes it possible to mix hot bitumen with cold and damp aggregates

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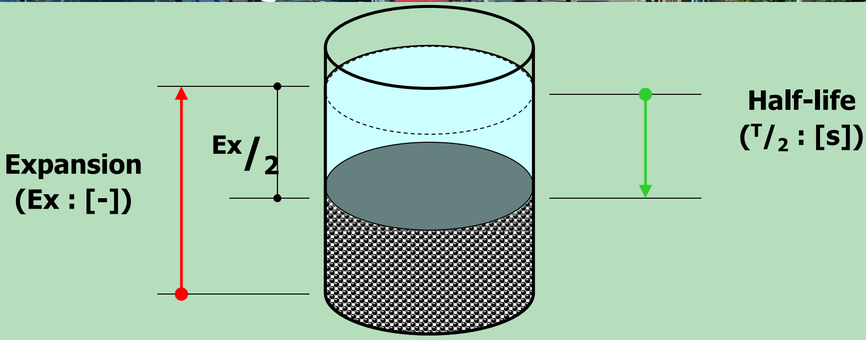
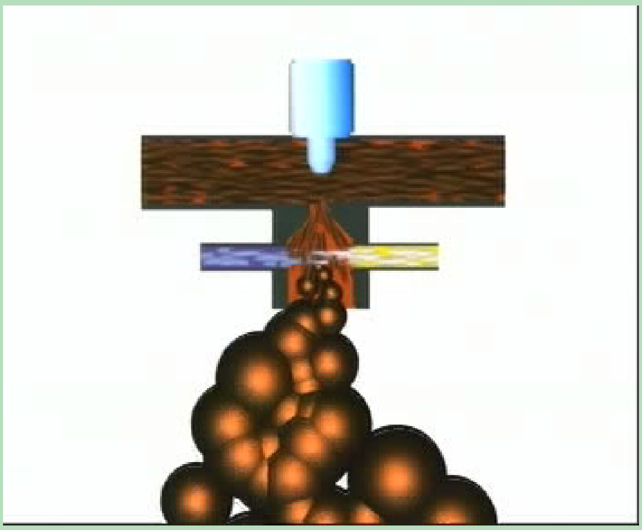
Characteristics of Foamed Bitumen



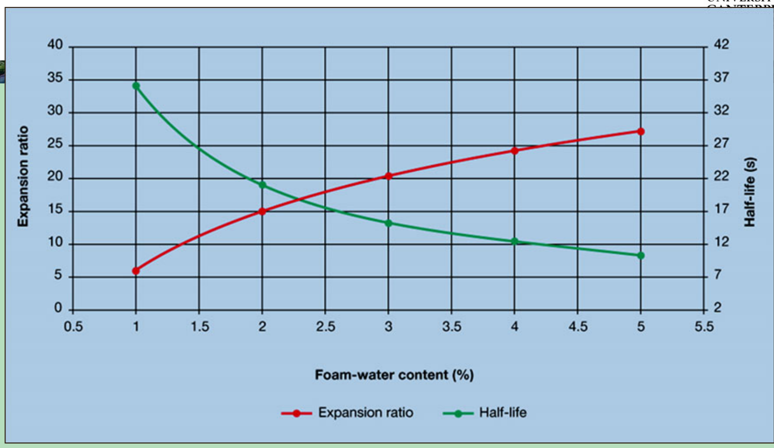
Quality of foamed bitumen: Expansion and Half-life

43



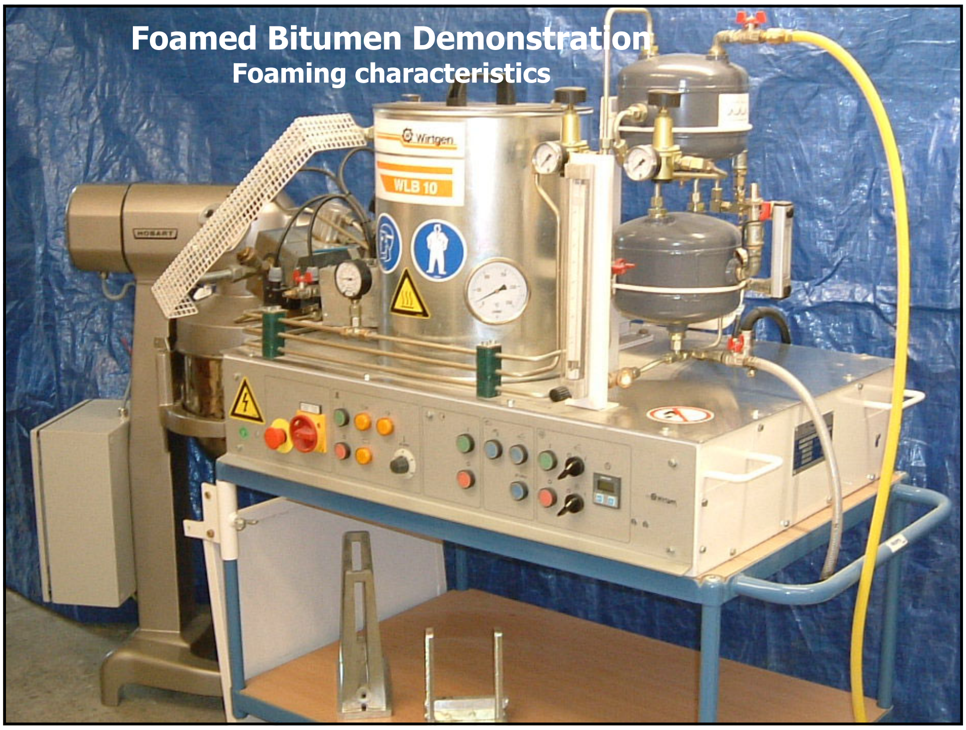
44

Characteristics of Foamed Bitumen



As the percentage of bitumen water is increased, the parameters "half-life" and "expansion" develop in opposite directions.

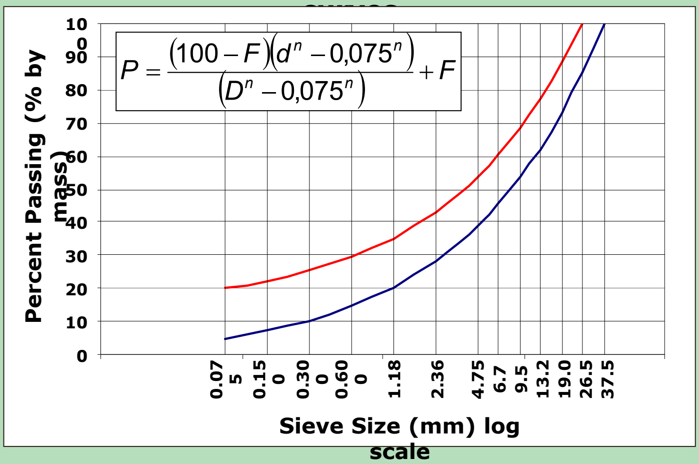
45



46

Foamed Bitumen Mix design

Characterisation of the sample in terms of:
- Particle Size Distribution and Plasticity (if any)
- Blending of aggregate (if necessary) to fit ideal



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Foamed Bitumen Mix design

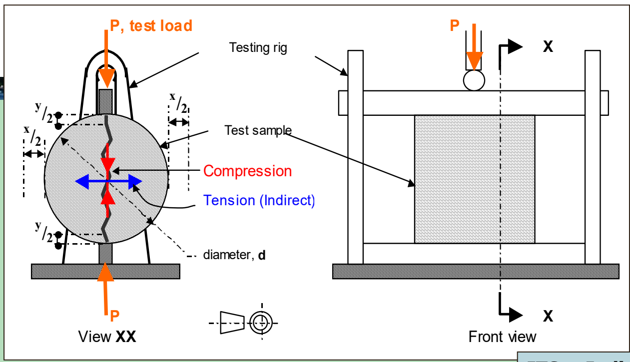
Determination of the Optimum Moisture Content (OMC) and Maximum Dry Density (MDD)

Mix design and manufacturing test samples:

- **Addition of cement (1.5 M.-%)**
- **Addition of water (at 80% - 90% of OMC)**
- **Addition of Foamed Bitumen (2; 3; 4; 5 M.-%): with WLB 10 directly into the laboratory mixer**
- **Production of briquettes by Marshall compaction – modified to 75 blows / briquette-side**
- **Store for 72 hours at 40 °C (to simulate 28 day strength)**
- **Indirect Tensile Strength (ITS) on dry and soaked briquettes at 25 °C**

48

Foamed Bitumen Mix design



$$ITS = \frac{2 * P}{\pi * d * h}$$

$$E_{SZ} = \frac{P * (0,274 + \mu)}{h * \Delta}$$

Indirect Tensile Strength by measuring the lateral strain the E-module can be calculated

ITS	= Indirect tensile strength	[N/mm²]
E_{SZ}	= Elasticity module	[N/mm²]
P	= Tensile breaking strength	[N]
d	= Briquette diameter	[mm]
h	= Briquette height	[mm]
μ	= Poissons ratio	[-]
Δ	= lateral deformation at break	[mm]

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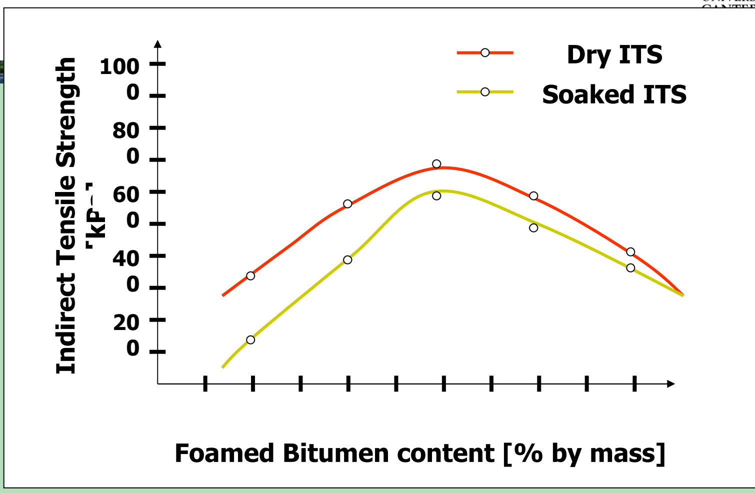
Foamed Bitumen Mix design



Optimisation of Foamed Bitumen content

50

Benefits of Foamed Bitumen Pavement Recycling

- **Operational**
  Continue during inclement weather
  Trafficked immediately after compaction
- **Environmental**
  Recycling of materials – sustainability of NZ resources
  Resource Management Act
- **Traffic Safety**
  Entire Recycling Train is accommodated within one traffic lane
  Dustless technology additional safety for personnel and traffic
- **Water Resistant base course**
  Bitumen coats particles up to 2 mm – water resistant and resistant to freeze / thaw cycles
- **Strong and Flexible base course**
  Load bearing by both bitumen rich mortar and particle interlock
  In addition bitumen rich mortar gives flexibility

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Save Construction Costs while Maintaining Strength

52

Durable

53

54


FHIS-3
GOBIERNO DE HONDURAS
PRESIDENCIA DE LA REPUBLICA
RECAP
UNIDAD EJECUTORA DE RECONSTRUCCION
DE EMERGENCIA DE CAMINOS Y PUENTES
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PROYECTO:
CONTRATO DE RECONSTRUCCION 005 R02
CARRETERAS Y OBRAS DE DRENAJE MENOR EN EL
DEPARTAMENTO DE YORO
SAN JOSE - EL NEGRITO
FUENTE DE FINANCIAMIENTO:
AGENCIA DE LOS ESTADOS UNIDOS PARA EL
DESARROLLO INTERNACIONAL (USAID) Y EL FONDO
HONDUREÑO DE INVERSION SOCIAL (FHIS)
CONVENIO DE DONACION No 522 0410
COSTO DEL PROYECTO: LPS 4166871.78
INICIO: 13 DE MARZO DEL 2000
CONTRATISTA: ETERNA S.A
LONGITUD 60.37 KM
DURACION 12 MESES
SUPERVISOR: LOUIS BERGER-ASP
Honduras

55


Japan

56

57

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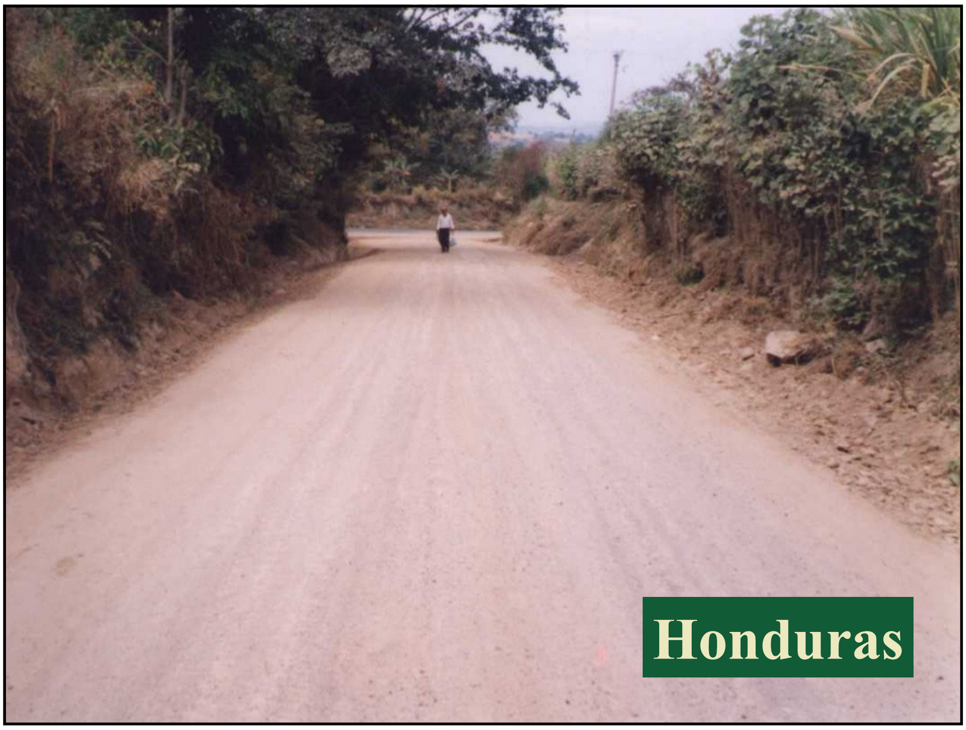
China
工 程 概 况
一.建设规模:
同皮线K51+135－K62
+977段,全长11.842公里.
二.技术标准:
按平原微丘区二级公路标
准设计:路面宽度9米,路基
宽度12米.
三.路面结构:
面层采用4cm沥青贯入,
基层采用23cm水泥稳定砂砾.
四.计划开.竣工日期:
2000年5月15日－9月20日.
五.施工单位:
普兰店市鹏宇公路工程处

61


Honduras

62

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